

The following rule allows established and related incoming connections:

```
sudo iptables -A INPUT -m conntrack --ctstate
ESTABLISHED,RELATED -j ACCEPT
```

This allows established outgoing connections:

```
sudo iptables -A OUTPUT -m conntrack --ctstate ESTABLISHED -j
ACCEPT
```

The following rule is for moving from local network to external network:

```
sudo iptables -A FORWARD -i eth1 -o eth0 -j ACCEPT
```

The following rule drops invalid packets:

```
sudo iptables -A INPUT -m conntrack --ctstate INVALID -j DROP
```

This blocks an IP address:

```
sudo iptables -A INPUT -s 192.168.122.1 -j DROP
```

This is for *Reject*:

```
sudo iptables -A INPUT -s 192.168.122.1 -j REJECT
```

This blocks connections to a network interface:

```
Iptables -A INPUT -i eth0 -s 15.15.15.51 -j DROP
```

This allows all incoming SSH:

```
sudo iptables -A OUTPUT -p tcp --dport 22 -m conntrack
--ctstate NEW,ESTABLISHED -j ACCEPT
sudo iptables -A INPUT -p tcp --sport 22 -m conntrack
--ctstate ESTABLISHED -j ACCEPT
```

Saving rules

Ubuntu: To save the Iptables rules in Ubuntu, use the

Iptables-persistent package. You can install it through apt-get.

```
sudo apt-get install Iptables-persistent
```

After updating your firewall rules, you can save the change by using the command given below:

```
sudo invoke-rc.d Iptables-persistent save
```

CentOS 6 and older: CentOS 6 and older versions of the OS use FirewallD by default. You can use Iptables service script to save the Iptables changes as shown below. Your current Iptables rules will be saved in the `/etc/sysconfig/Iptables` file.

```
sudo service Iptables save
```

This article has covered the basics of Iptables -- what it is, how it works, the different types of Iptables, options to Iptables and a basic set of rules. However, a lot more can be done with Iptables. You can mix and match the different sets of rules as per your need. 

References

- [1] https://access.redhat.com/documentation/en-US/Red_Hat_Enterprise_Linux/6/html-single/Security_Guide/index.html#sect-Security_Guide-Firewalls-FORWARD_and_NAT_Rules
- [2] <http://searchenterpriselinix.techtarget.com/definition/iptables>
- [3] <http://opensourceforu.com/2015/04/iptables-the-default-linux-firewall/>
- [4] <https://n0where.net/how-does-it-work-iptables/>
- [5] <https://wiki.archlinux.org/index.php/iptables>
- [6] <http://www.thegeekstuff.com/2011/01/iptables-fundamentals/>
- [7] <https://www.frozentux.net/iptables-tutorial/iptables-tutorial.html#BASICS>
- [8] <https://www.digitalocean.com/community/tutorials/iptables-essentials-common-firewall-rules-and-commands>

By: Roopendra Vishwakarma

The author has over five years of experience in open source technology including NoSQL, search engines and Web development. He can be reached at roopendra@techieroop.com.

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